A · Axioms act as directions

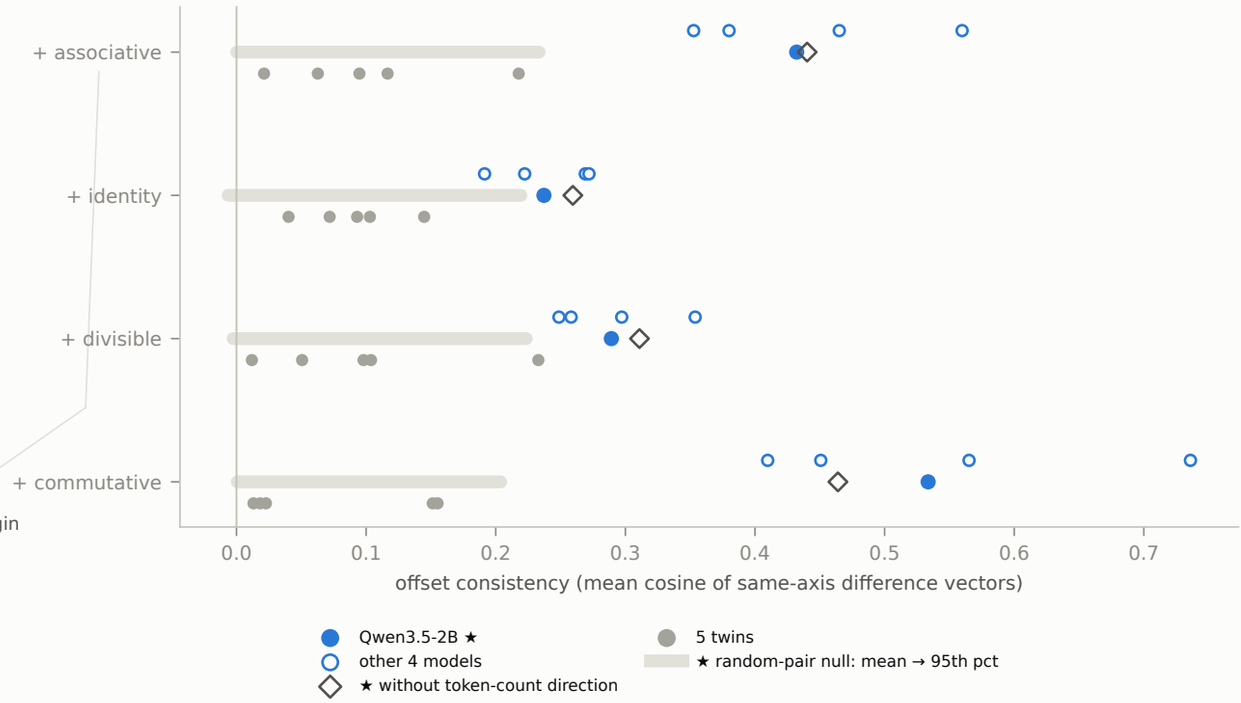
Qwen3.5-2B ★

'+ commutativity' arrows from one origin
consistency 0.53

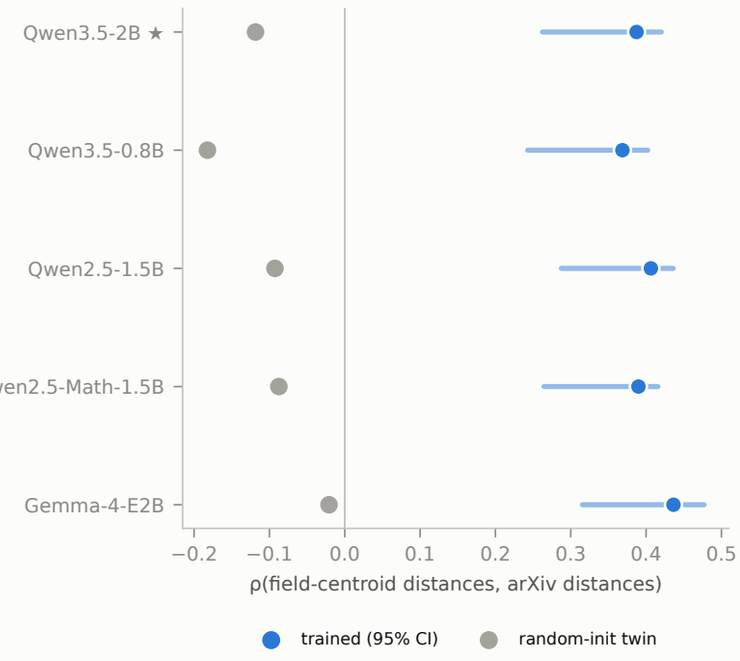
its random-init twin

'+ commutativity' arrows from one origin
consistency 0.01

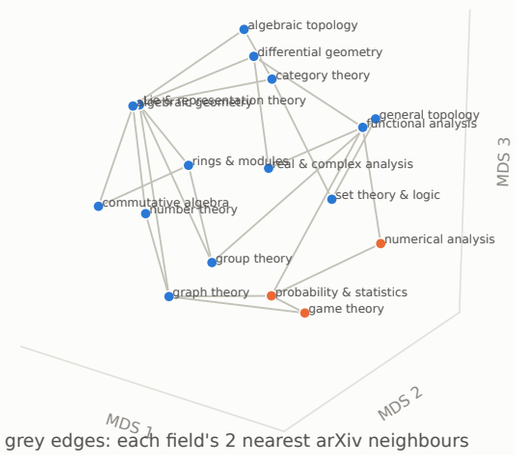
B · All four axioms, all five models: more parallel than the twins (clearest for associativity and commutativity)



C · Fields form a map in every model

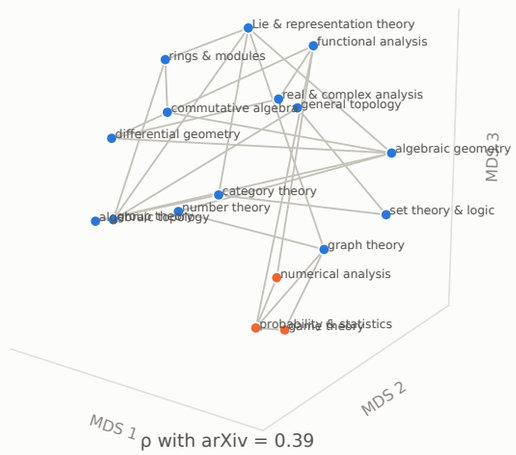


D · arXiv cross-listing (ground truth)



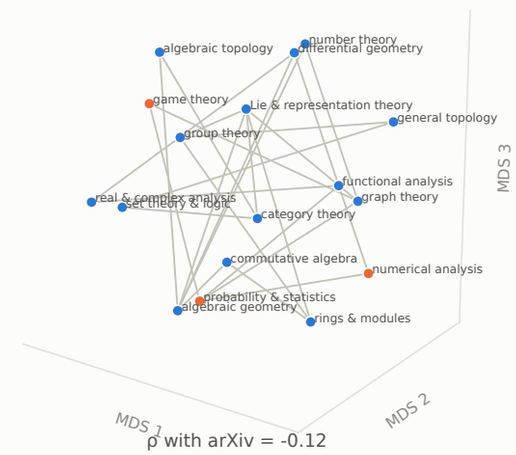
grey edges: each field's 2 nearest arXiv neighbours

Qwen3.5-2B ★: related fields sit together



ρ with arXiv = 0.39

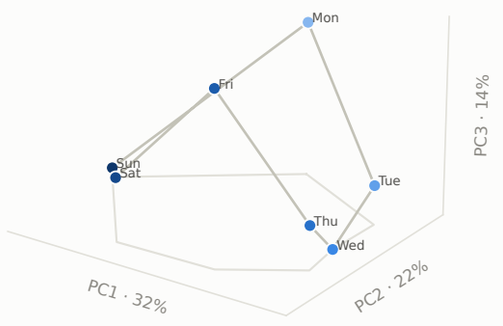
Its random-init twin: no map



ρ with arXiv = -0.12

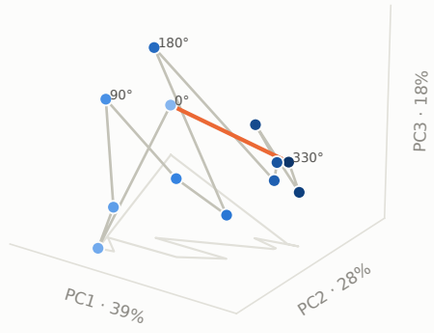
pure field
applied field

E · Days: a circle (positive control)



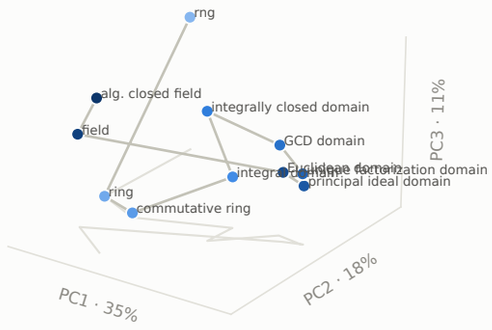
cycle score 1.00 (twin 0.14)

Angles: not a circle



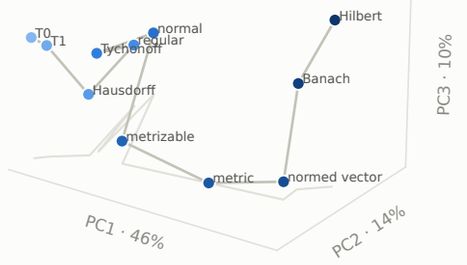
d(0°, 330°) = 2.11 × neighbour step (twin 0.80)

Ring chain: an ordered path



Robinson 0.52 (twin 0.30)

Space chain: an ordered path



Robinson 0.65 (twin 0.33)